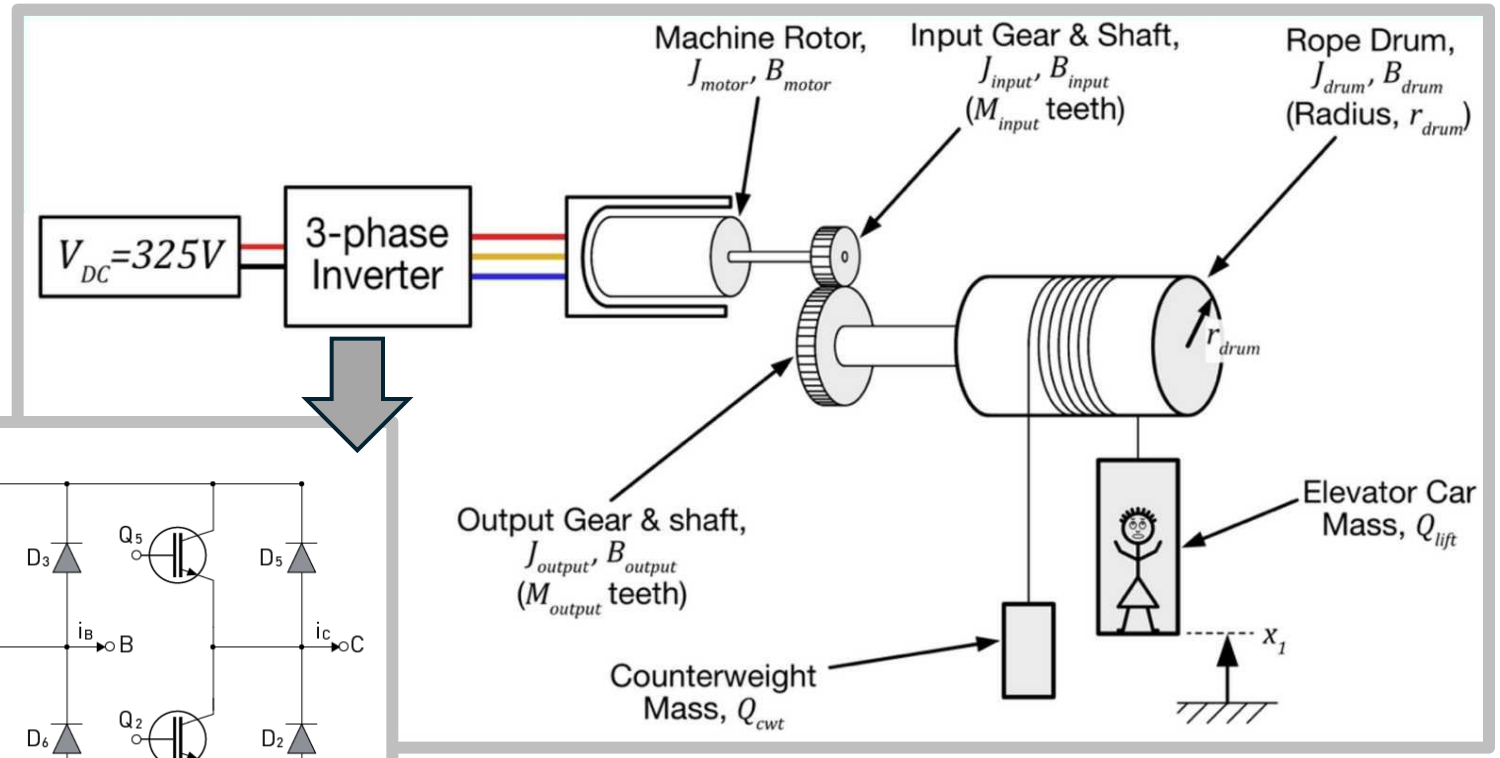
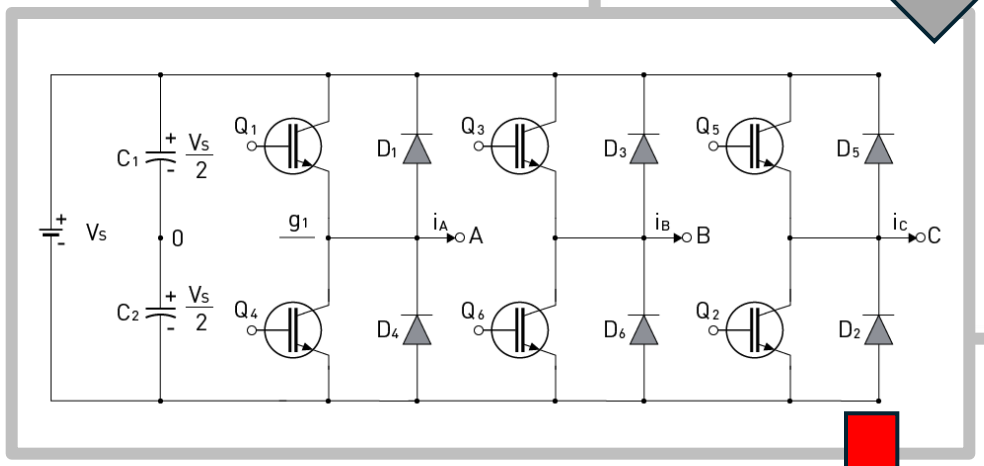
From System to Device: A Stepwise Zoom-In

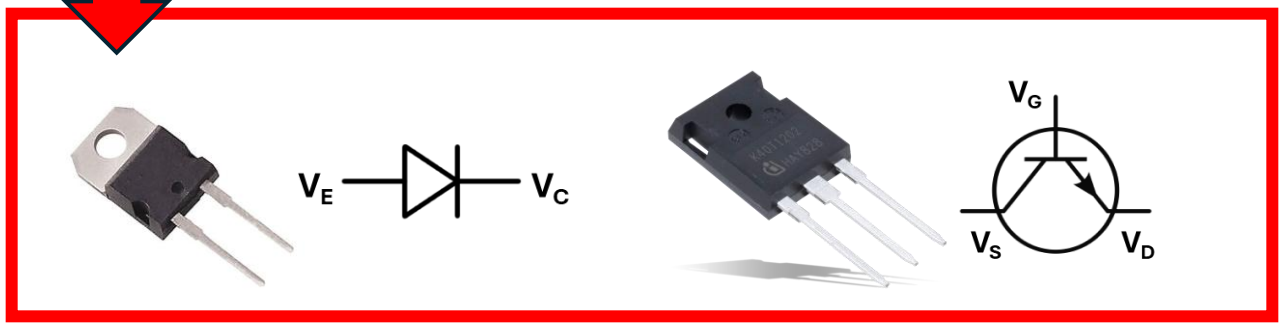
Level (Topic) 1-Electric Machine System (Machine and load)



Level (Topic) 2-Inverter Circuit (PWM and modulation)



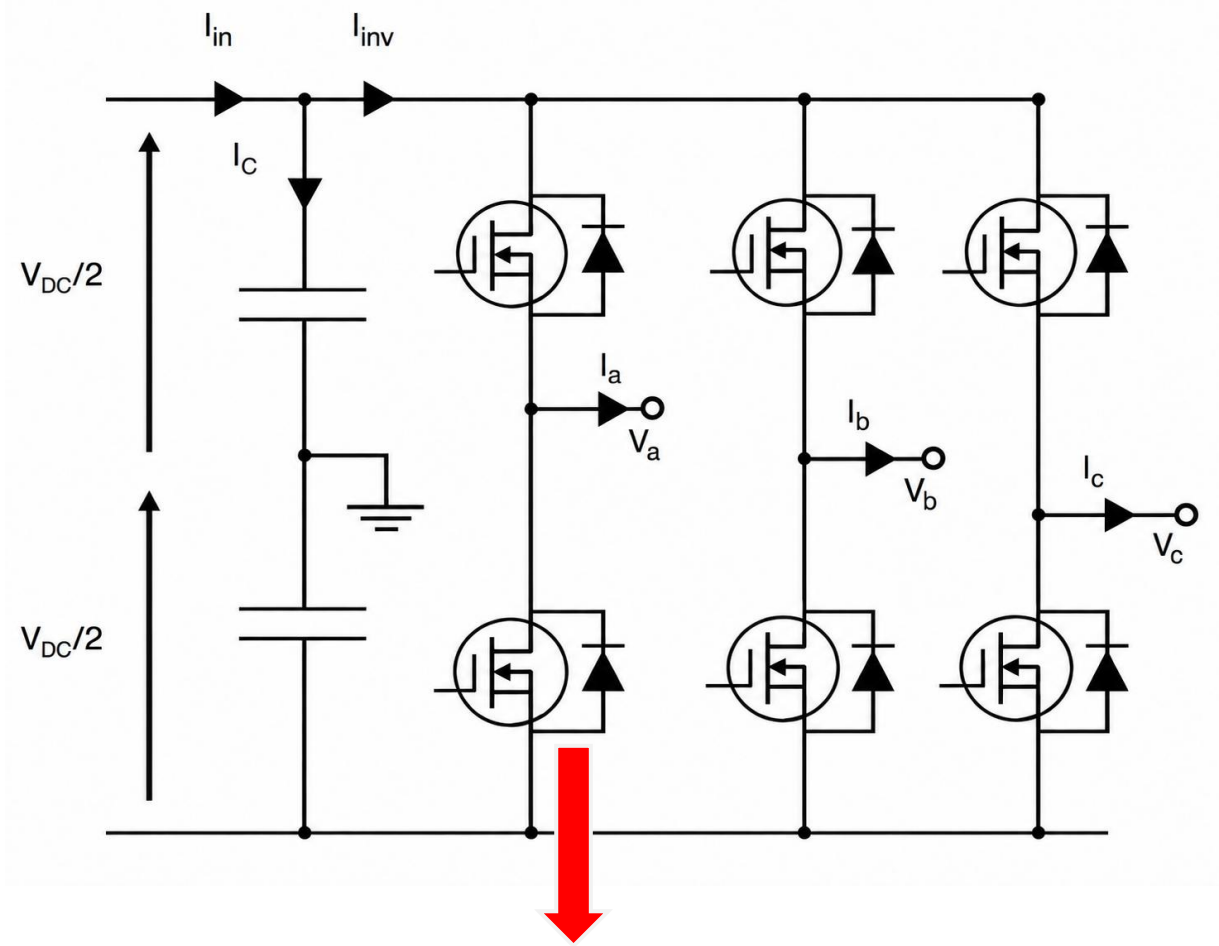
Level (Topic) 3-Power Electronic Components (conduction and switching losses)

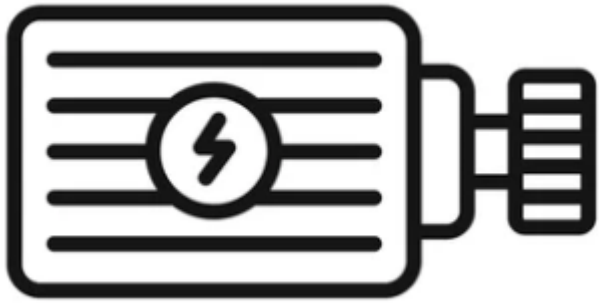


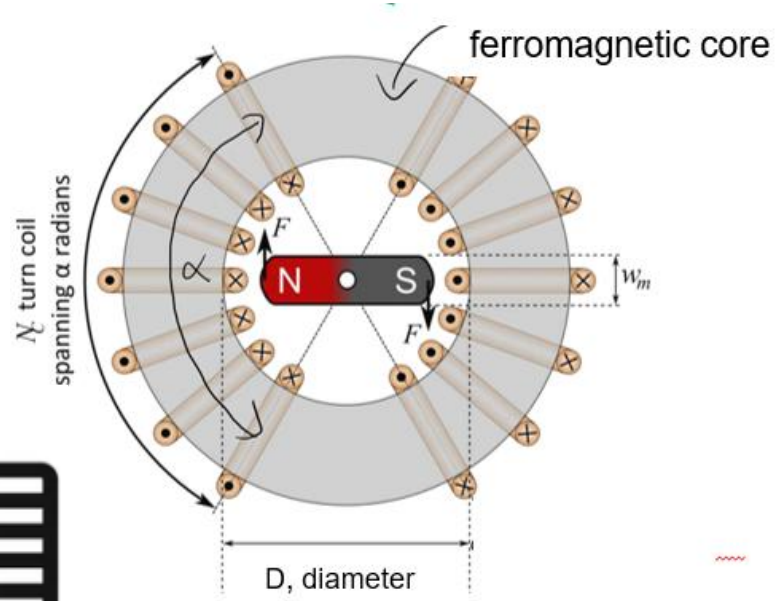
How do PWM decisions affect losses in the inverter switches, diodes, and motor?

From PWM to Inverter Losses

How do modulation choices affect losses in a motor drive?



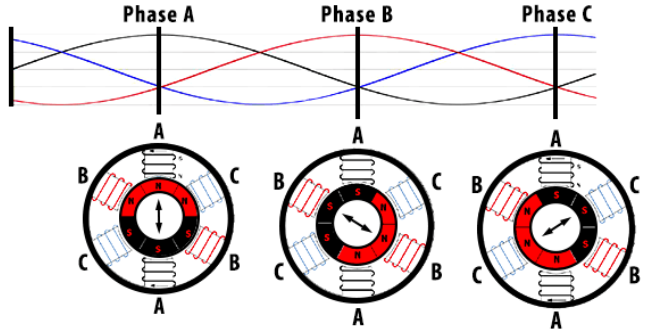




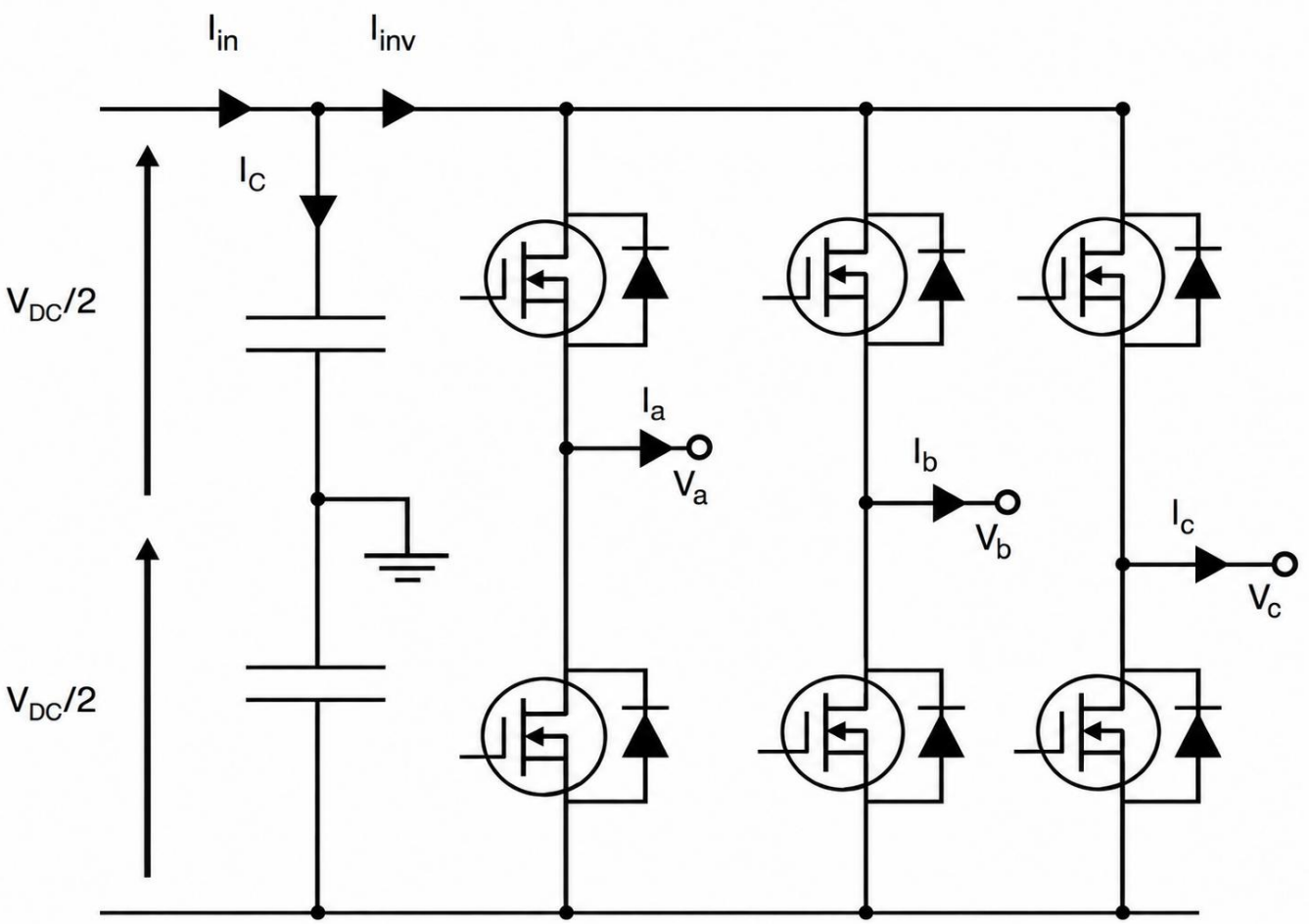
Conduction losses
Switching losses
Reverse losses

Harmonic losses

3 PHASE INDUCTION MOTOR



The Three-Phase Two-Level Inverter We Will Analyse



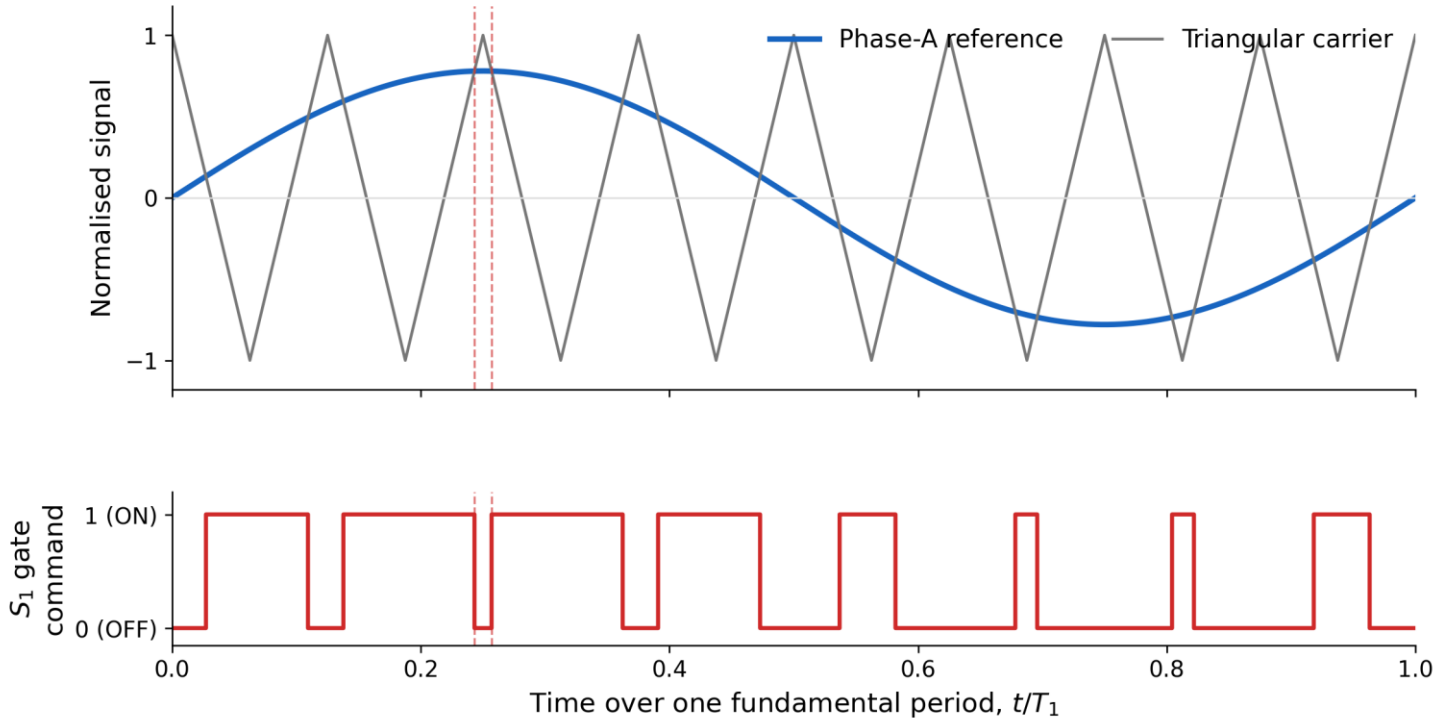
Gate command — when does S1 switch?

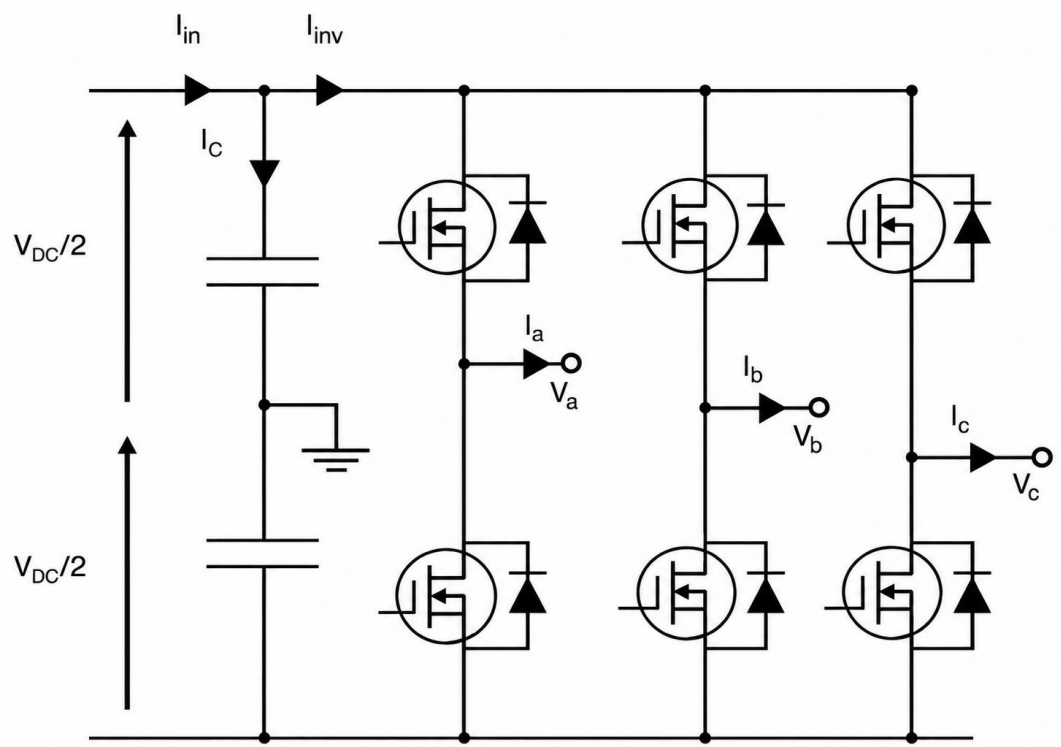
Switch voltage and current — what happens during a transition?

Phase current I_a — how much current is flowing at that moment?

We will examine one phase leg first, then return to the complete three-phase drive.

PWM Determines When the Switches Turn On and Off

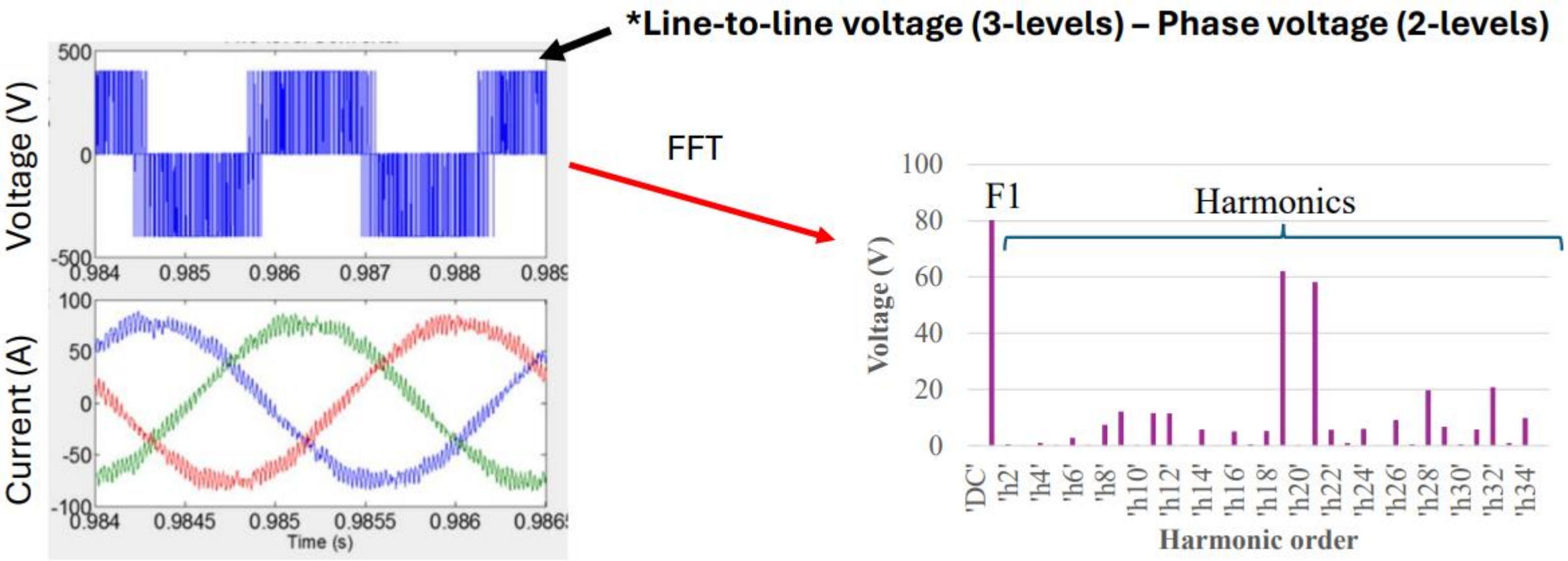




Gate command for the upper switch S1
ON (1): sinusoidal reference > triangular carrier
OFF (0): sinusoidal reference < triangular carrier
Each crossing changes the switching command, producing a PWM pulse train.

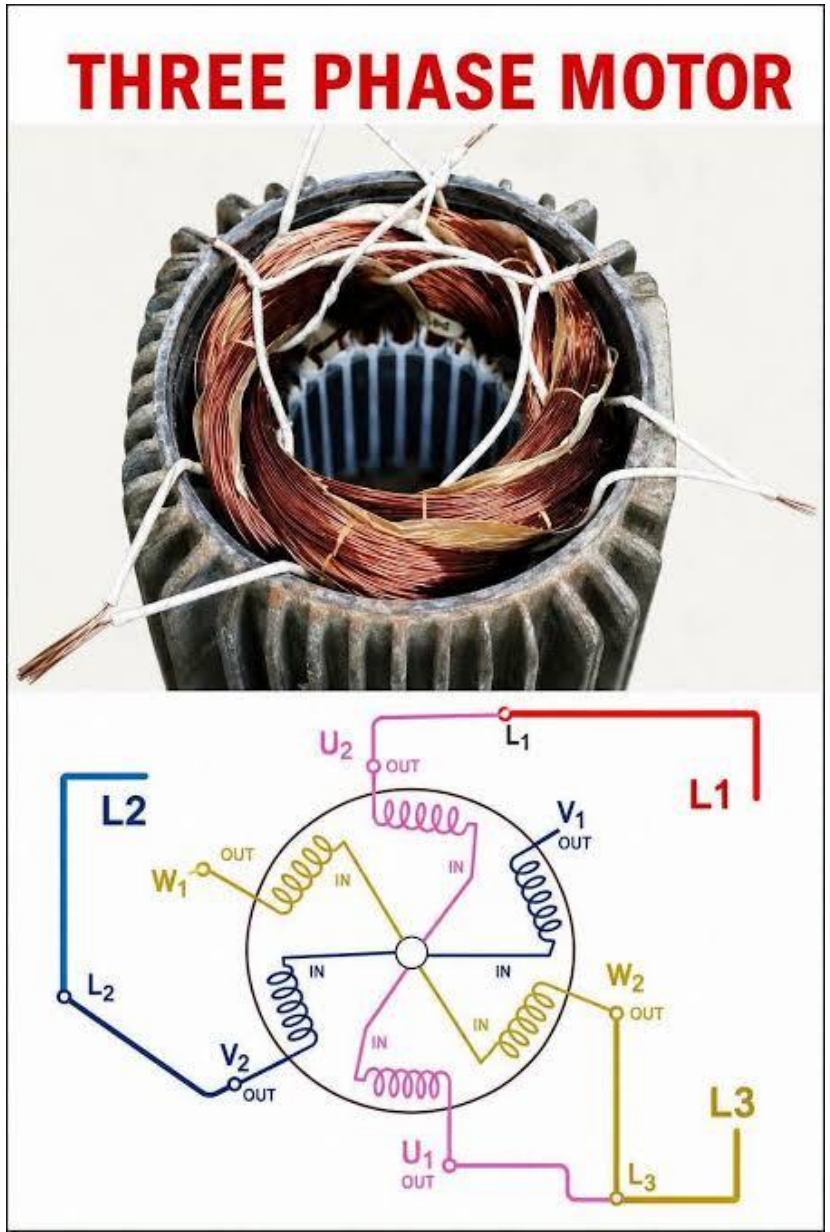
These switching commands create the inverter output voltage—and its harmonics.

PWM Produces the Fundamental Voltage—and Harmonics



- PWM produces the required fundamental voltage.
- Switching also introduces harmonics around the carrier frequency and its multiples.

Motor Winding Copper Loss: Fundamental and Harmonic Contributions

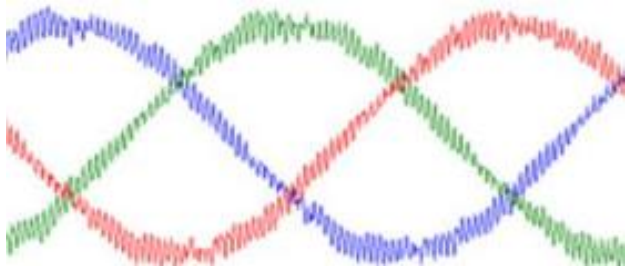


$$P_{\text{cu,total}} = P_{\text{cu,1}} + P_{\text{cu,h}}$$

$$P_{\text{cu,1}} = 3 r_s I_{1,\text{rms}}^2$$

$$I_{1,\text{rms}} = \frac{I_{1,\text{peak}}}{\sqrt{2}}$$

$$P_{\text{cu,h}} = 3 r_s \sum_{n>1} I_{n,\text{rms}}^2$$



- Switching ripple contains harmonic currents
- These currents flow through the winding resistance and produce heat

Harmonic Currents Cause Additional Motor Copper Losses

- The motor impedance determines the resulting harmonic currents.

$$I_{n,\mathrm{rms}} \approx \frac{V_{n,\mathrm{rms}}}{|Z|}$$

- r_s : resistance of one phase winding
- Z : impedance
- I_{n_rms} : RMS value of the *n*-th harmonic current
- fn

$$|Z| = \sqrt{r_s^2 + (2\pi f_n L_s)^2}$$

$$I_{n,\mathrm{rms}} \approx \frac{V_{n,\mathrm{rms}}}{\sqrt{r_s^2 + (2\pi f_n L_s)^2}}$$

$$(2\pi f_n L_s \gg r_s)$$

$$I_{n,\mathrm{rms}} \approx \frac{V_{n,\mathrm{rms}}}{2\pi f_n L_s}$$

$$f_n \approx k f_{\mathrm{sw}} \pm m f_1$$

$$I_{n,\mathrm{rms}} \approx \frac{V_{n,\mathrm{rms}}}{2\pi (k f_{sw} \pm m f_1) L_s}$$

$$I_{n,\mathrm{rms}} \approx \frac{V_{n,\mathrm{rms}}}{2\pi (k f_{sw}) L_s}$$

$$(f_{\mathrm{sw}} \gg f_1)$$

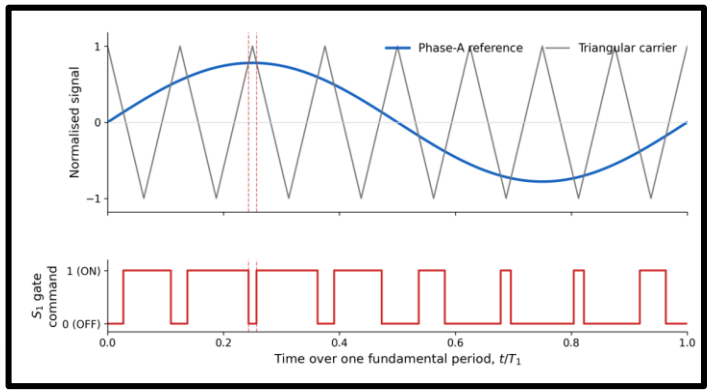
$f_{\mathrm{sw}} = 20\ \mathrm{kHz}$

$f_1 = 50\ \mathrm{Hz}$

PWM Affects Losses in Both the Motor and the Inverter

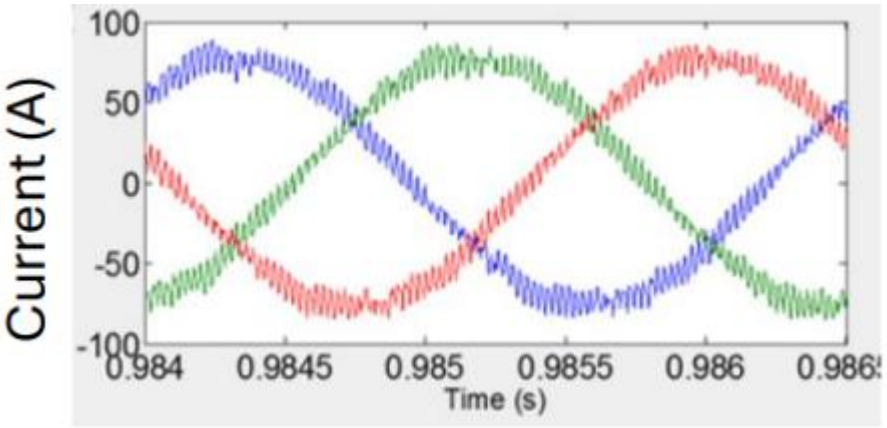
Output voltage and current

Device conduction and switching



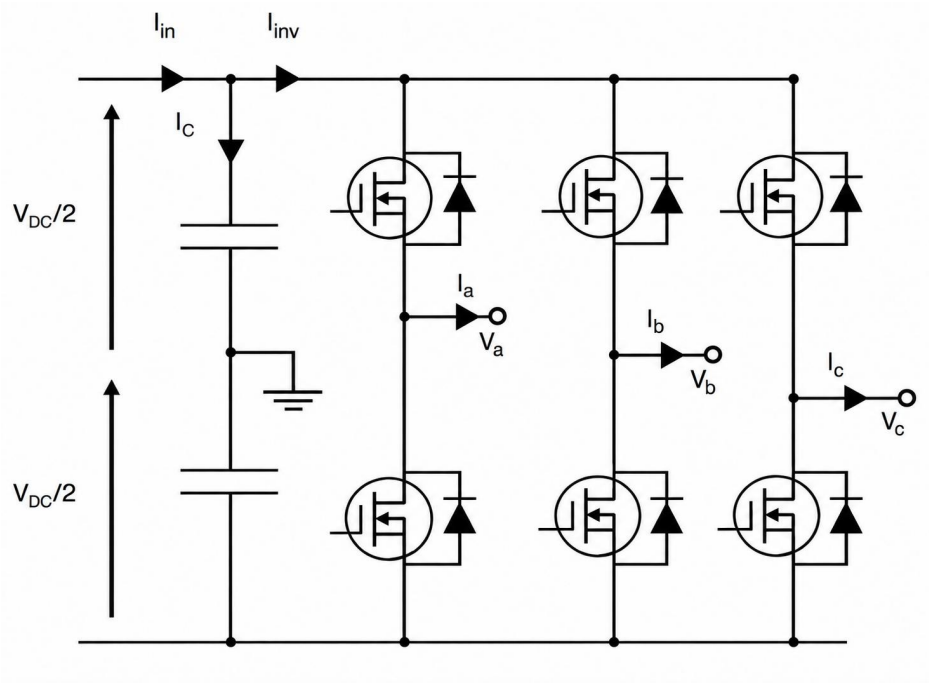
PWM

Motor-side losses



- Voltage harmonics produce harmonic currents.
- These currents cause additional winding copper losses.

Inverter-side losses

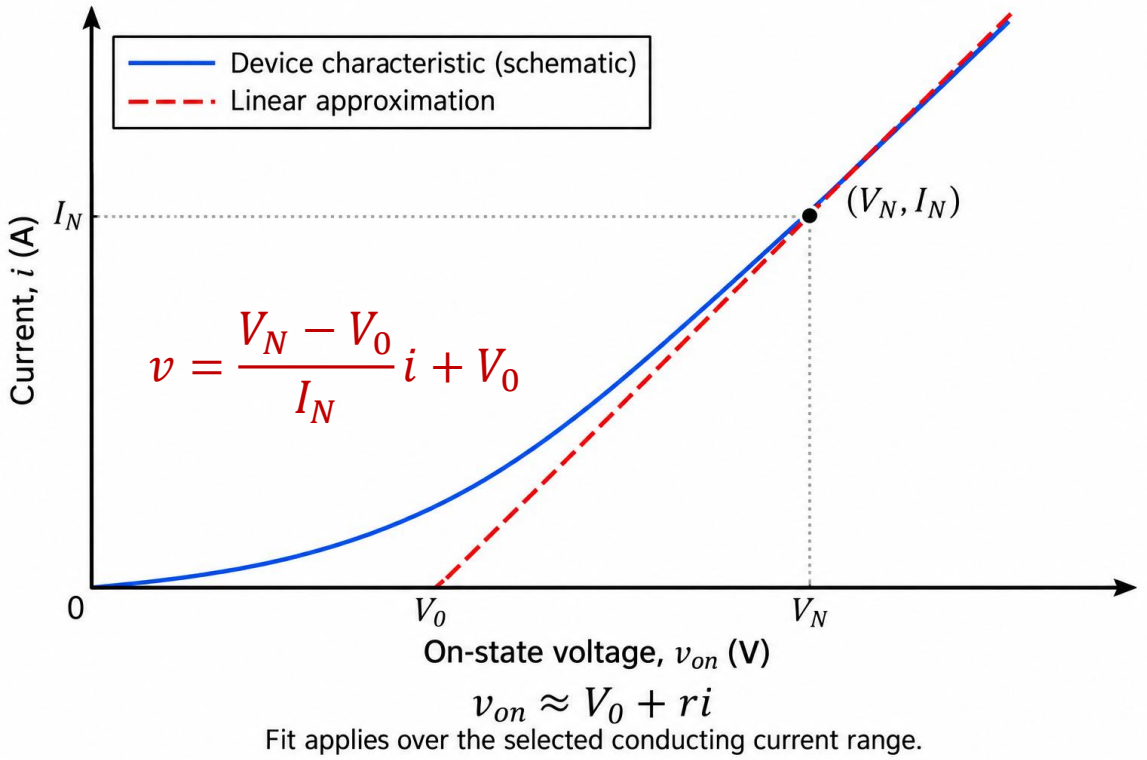


- Real devices dissipate power during conduction and switching.

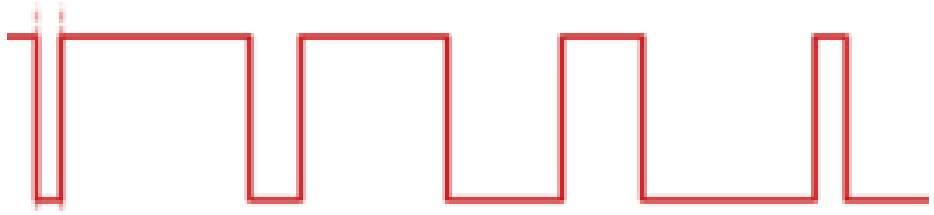
What happens inside a switching device during each PWM transition?

Linear Approximation for Conduction Loss

$$p=vi$$



What is missing?



$$\bar{p}_{\text{cond}} = \left(\frac{V_N - V_0}{I_N} i + V_0\right) i\,\delta$$

$$p = \left(\frac{V_N - V_0}{I_N} i + V_0\right) i$$

Can we express current and duty ratio as functions of angle α?

Expressing Conduction Loss as a Function of Angle α

$$\bar{p}_{\text{cond}} \approx \left(\frac{V_N - V_0}{I_N} i^2 + V_0 i \right) \delta$$

For i:

$$i = \hat{I} \sin \alpha$$

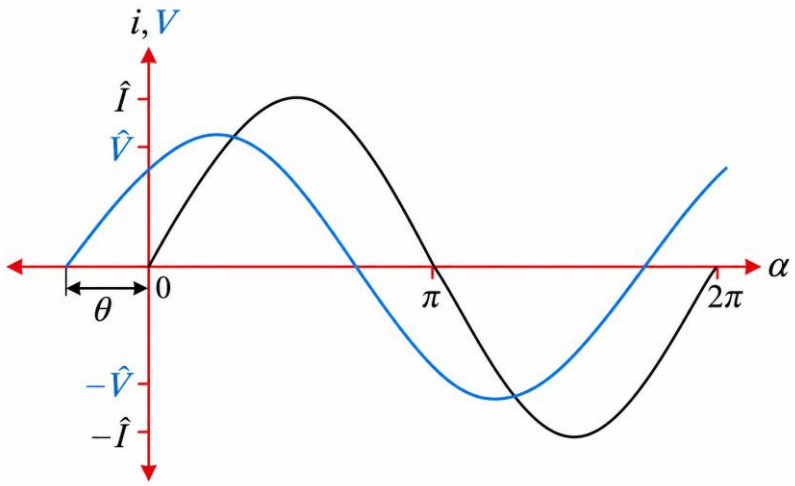
$$\hat{I} = \sqrt{2}\, I_{\text{rms}}$$

For δ:

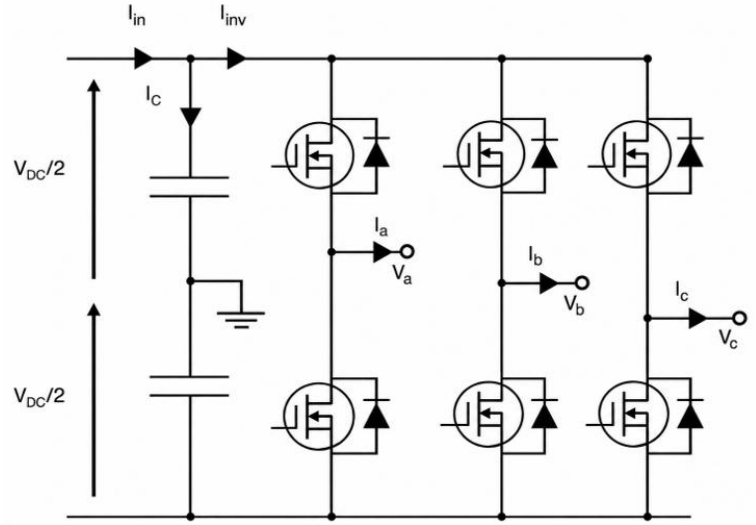
$$\delta = M \sin \alpha$$

$$M = \frac{\hat{V}_{\text{ref}}}{\hat{V}_c}$$

Phase difference



Voltage reference

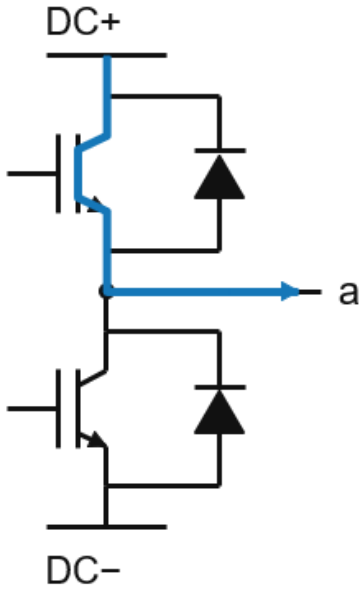


$$\delta = M \sin(\alpha + \theta)$$

$$\delta(\alpha) = \frac{1}{2}[1 + M \sin(\alpha + \theta)]$$

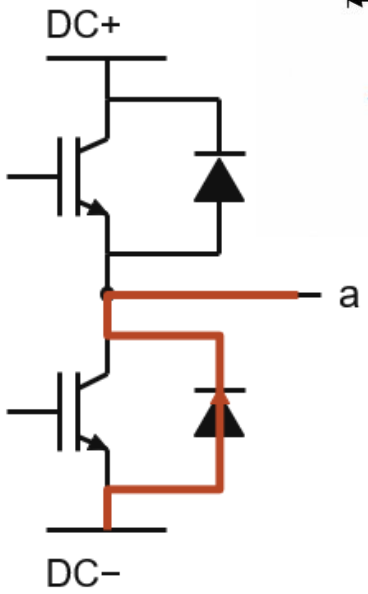
Current Paths and Conduction Intervals

Positive current: a to motor



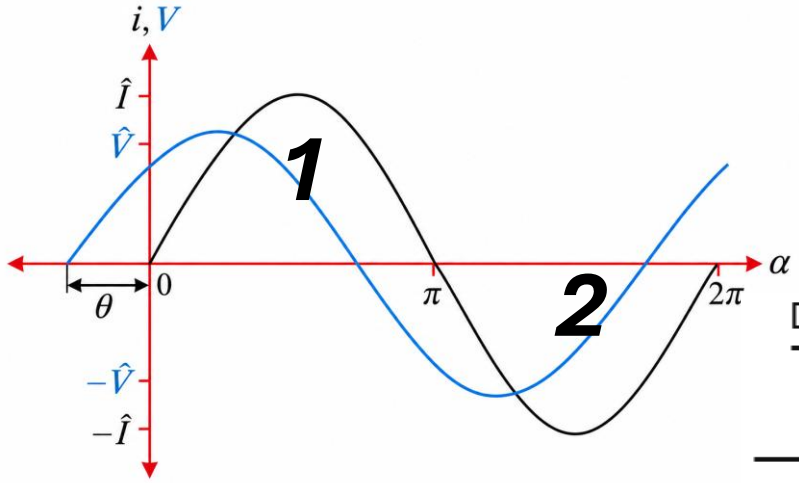
S_1 carries current

Duration: δT

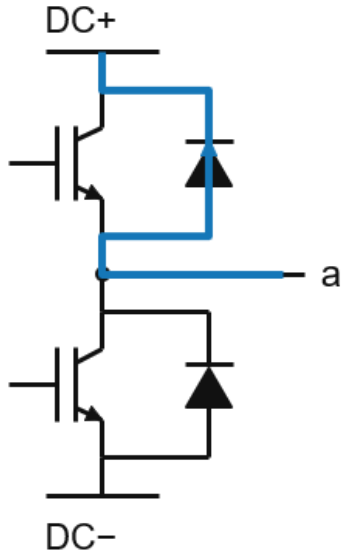


D_2 carries current

Duration: $(1-\delta)T$

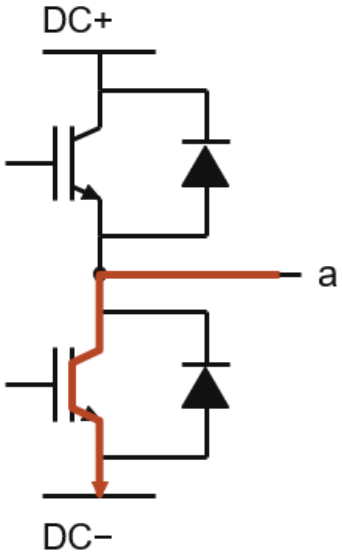


Positive current: motor to a



D_1 carries current

Duration: δT

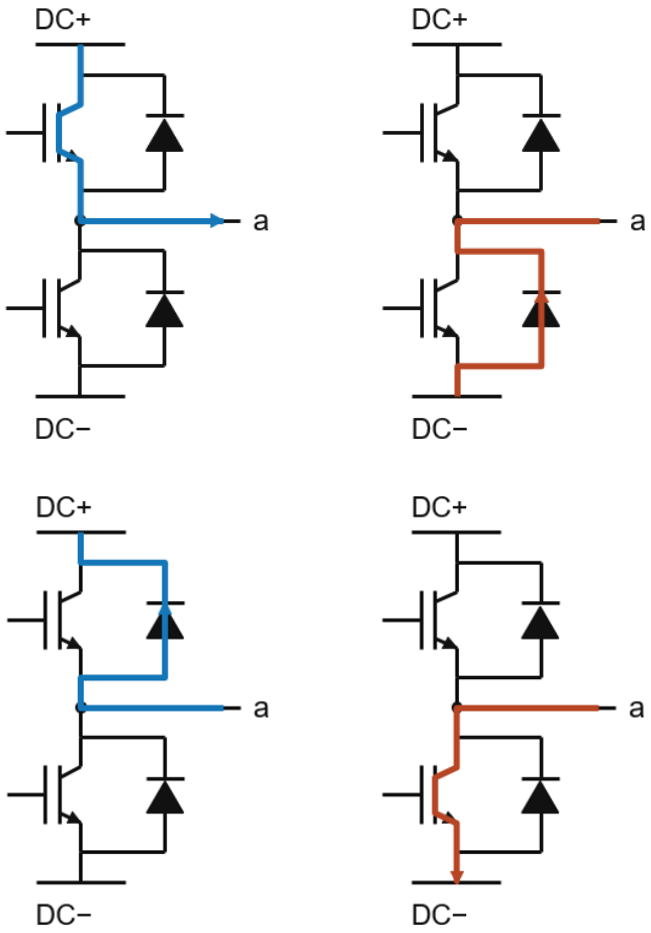


S_2 carries current

Duration: $(1-\delta)T$

IGBTs conduct forward current. Antiparallel diodes carry reverse current.

Combined Conduction Loss of Both IGBTs over the Cycle



$$\bar{P} = \frac{1}{2\pi}\int_0^{\pi}\left(\frac{V_{N(S_1)} - V_{0(S_1)}}{I_{N(S_1)}}\hat{I}^2\sin^2\alpha + V_{0(S_1)}\hat{I}\sin\alpha\right)\frac{1}{2}(1 + M\sin(\alpha+\theta))\,d\alpha$$

$$\bar{P} = \frac{1}{2\pi}\int_{\pi}^{2\pi}\left(\frac{V_{N(S_2)} - V_{0(S_2)}}{I_{N(S_2)}}\hat{I}^2\sin^2\alpha - V_{0(S_2)}\hat{I}\sin\alpha\right)\frac{1}{2}(1 - M\sin(\alpha+\theta))\,d\alpha$$

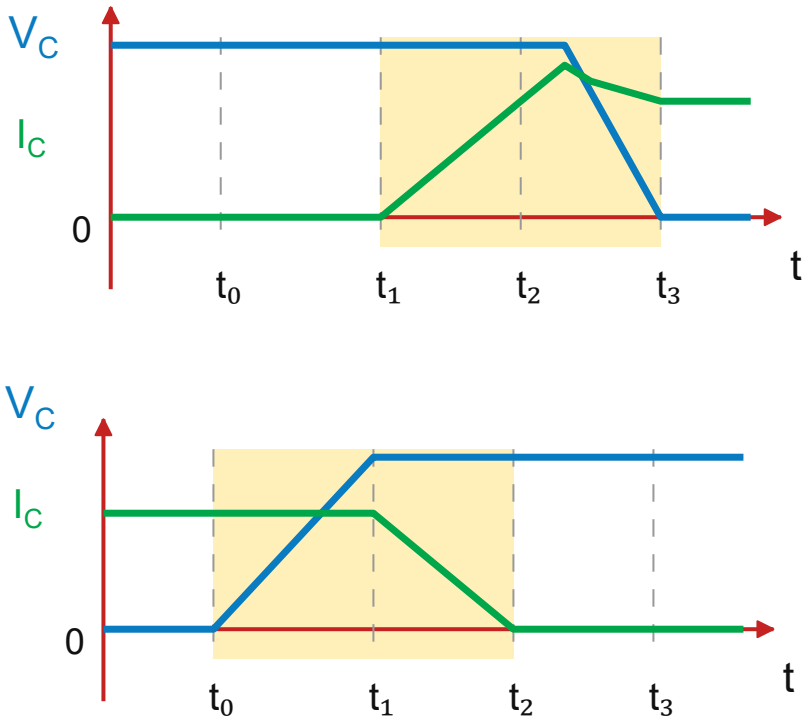
Diode conduction losses can be calculated similarly; the derivation is omitted here.

$$\bar{P}_{\text{cond},D_2} = \frac{1}{2\pi}\int_0^{\pi} \mathrm{V_F}\,(\alpha)\,\hat{I}\sin\alpha\,\frac{1}{2}[1 - M\sin(\alpha+\theta)]\,d\alpha$$

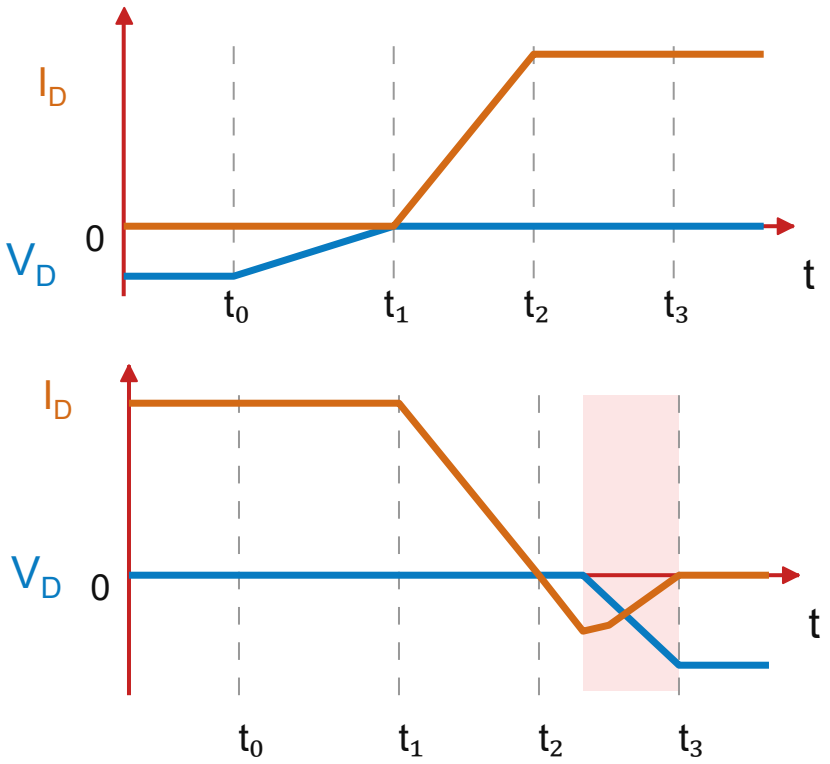
$$v_F(\alpha) \approx V_{0,D} + r_D\hat{I}\sin\alpha$$

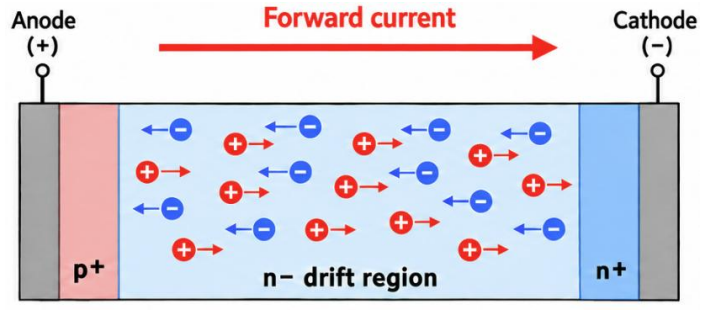
Switching Losses in Transistors and Diodes

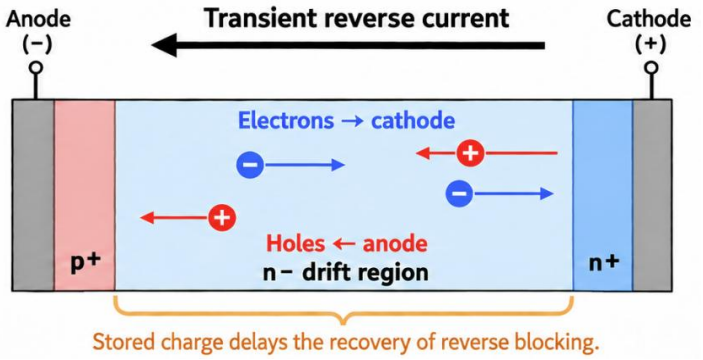
Voltage and transistor current



Voltage and current



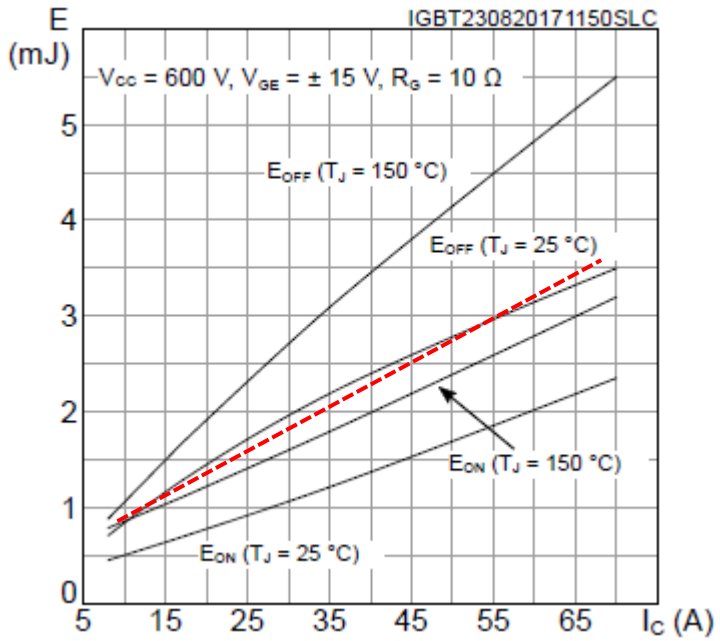




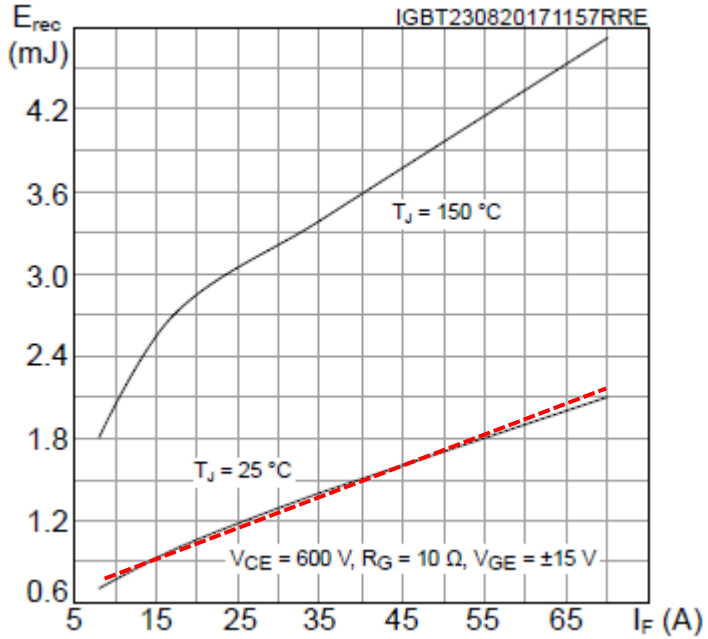
Device	Transition	First	Then
IGBT	Turn-on	Current rises	Voltage falls
IGBT	Turn-off	Voltage rises	Current falls
Diode	Turn-on	Voltage falls	Current rises
Diode	Turn-off	Forward current falls through zero and becomes negative	Reverse voltage rises, then reverse current decays to zero

From Datasheet Energy Curves to Average Power Loss

For IGBT (switching loss)



For diode (reverse loss)



1. Fit energy curves:

$$E_{\mathrm{sw}(x)} = \frac{v_{\mathrm{sw}}}{V_{\mathrm{DD}}}\left(A_{(x)} + B_{(x)} i_{\mathrm{sw}} + C_{(x)} i_{\mathrm{sw}}^2\right), \qquad \text{where } x = \text{on, off, rev}$$

2. Express in α:

$$E_{\mathrm{sw}(x)} = \frac{V_{\mathrm{sw}}}{V_{\mathrm{DD}}}\left(A_{(x)} + B_{(x)} \hat{I} \sin\alpha + C_{(x)} \hat{I}^2 \sin^2\alpha\right)$$

3. Average power loss:

$$\bar{P}_{(x)} = \frac{f_s}{2\pi} \int_0^{\pi} \frac{V_{\mathrm{DC}}}{V_{\mathrm{DD}}}\left(A_{(x)} + B_{(x)} \hat{I} \sin\alpha + C_{(x)} \hat{I}^2 \sin^2\alpha\right) d\alpha$$

Summary of IGBT and Diode Losses

For IGBT, the conduction loss:

$$\bar{P} = \frac{1}{2\pi}\int_0^{\pi}\left(\frac{V_{N(S_1)} - V_{0(S_1)}}{I_{N(S_1)}}\hat{I}^2\sin^2\alpha + V_{0(S_1)}\hat{I}\sin\alpha\right)\frac{1}{2}(1 + M\sin(\alpha+\theta))\,d\alpha$$

For diodes, the conduction loss:

$$\bar{P}_{\text{cond},D_2} = \frac{1}{2\pi}\int_0^{\pi}(V_{0,D} + r_D\hat{I}\sin\alpha)\;\hat{I}\sin\alpha\;\frac{1}{2}[1 - M\sin(\alpha+\theta)]\,d\alpha$$

How Long

For IGBT, the switching loss:

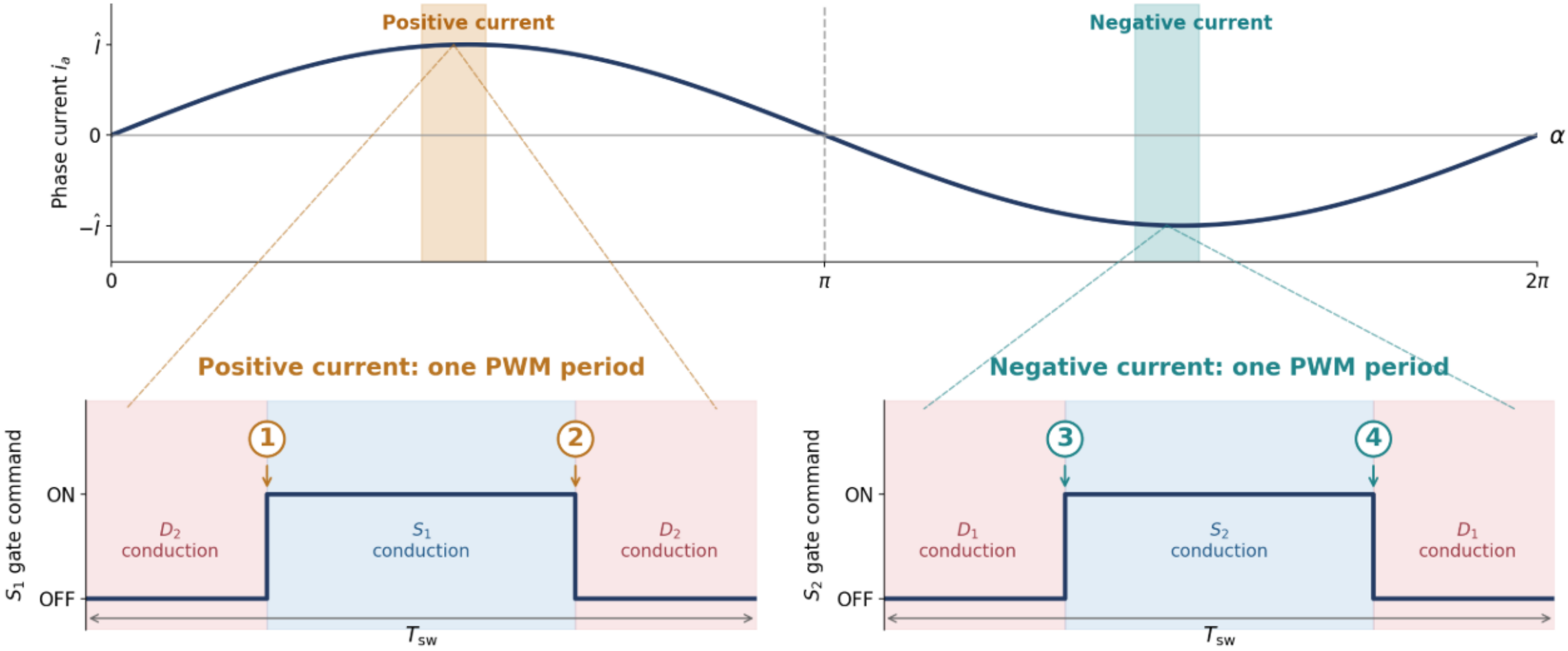
$$\bar{P}_{(x)} = \frac{f_s}{2\pi}\int_0^{\pi}\frac{V_{\text{DC}}}{V_{\text{DD}}}\left(A_{(x)} + B_{(x)}\hat{I}\sin\alpha + C_{(x)}\hat{I}^2\sin^2\alpha\right)d\alpha$$

For diodes, the reverse loss:

$$\bar{P}_{(x)} = \frac{f_s}{2\pi}\int_0^{\pi}\frac{V_{\text{DC}}}{V_{\text{DD}}}\left(A_{(x)} + B_{(x)}\hat{I}\sin\alpha + C_{(x)}\hat{I}^2\sin^2\alpha\right)d\alpha$$

How Often

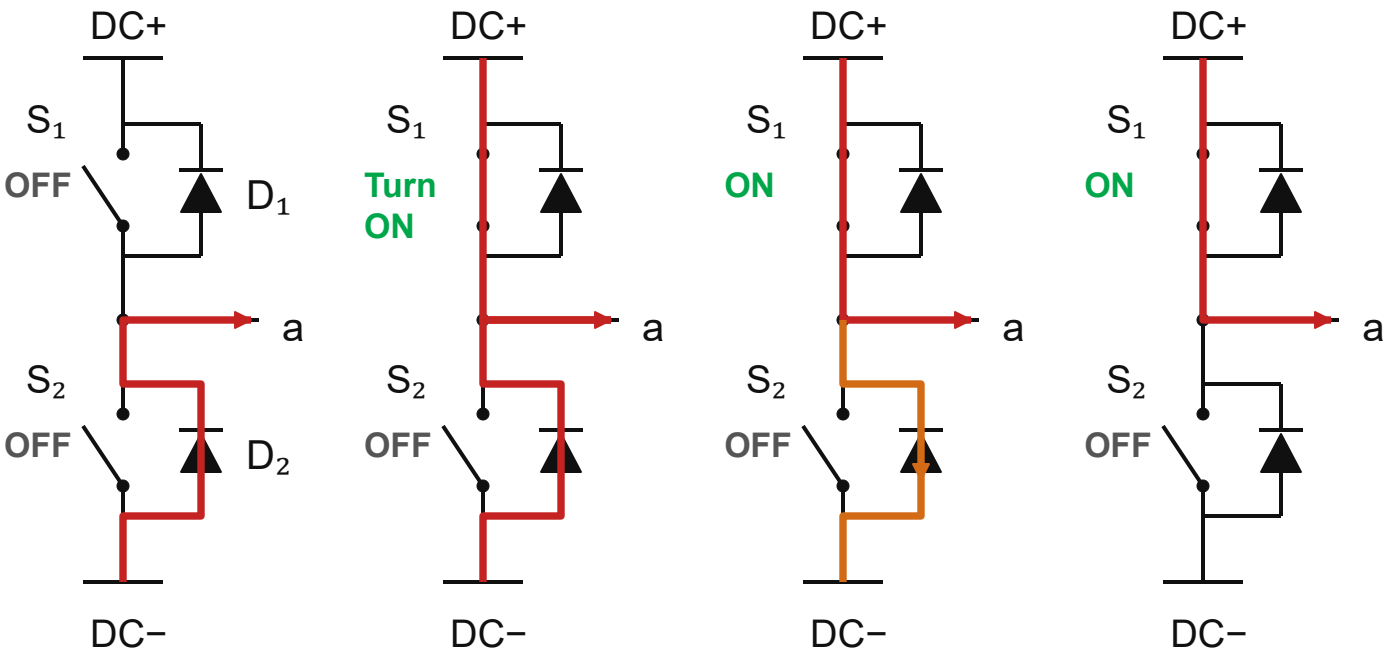
Four Switching Cases within a Fundamental Cycle



S_1 turn-on with positive phase current

$0 < \alpha < \pi \quad i_a > 0$

Ideal IGBT switches. Red: load path. Orange: recovery current.



$t < t_1$

D_2 freewheels

t_1 to t_2

S_1 takes current

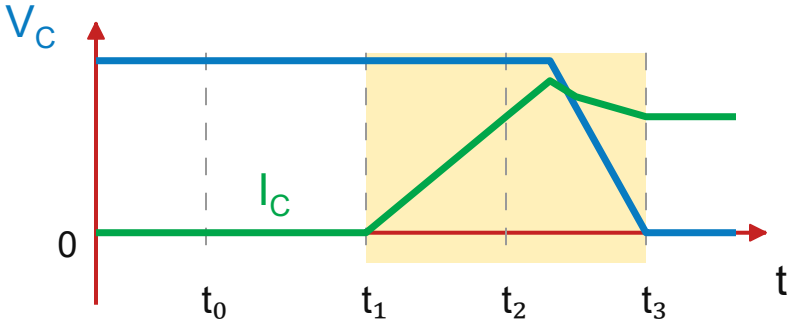
t_2 to t_3

D_2 recovers

$t > t_3$

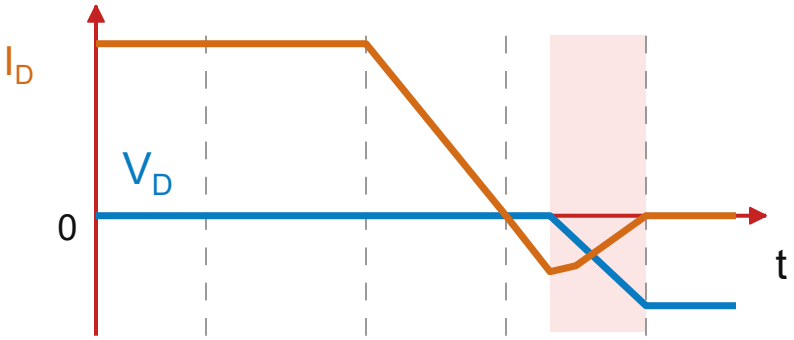
S_1 conducts

S1 voltage and transistor current



Shaded: turn-on overlap

D2 voltage and current



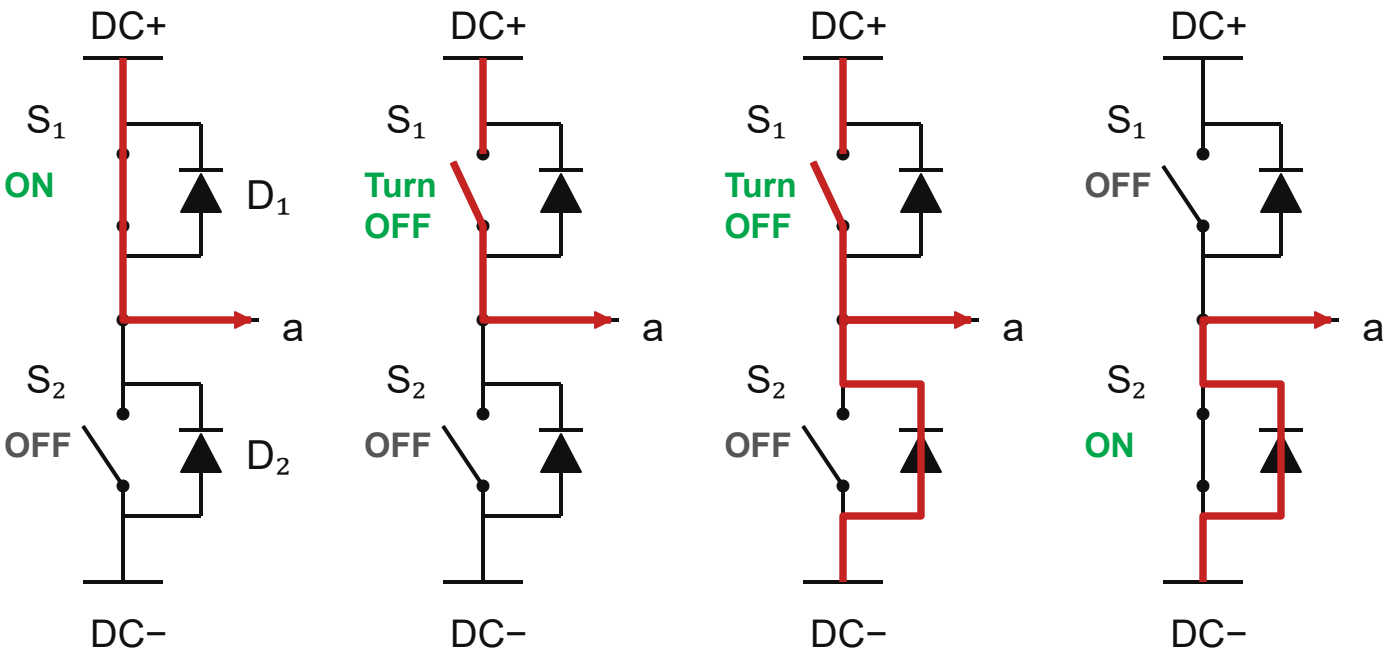
Shaded: diode voltage–current overlap

S_1 turns on and forces D_2 into reverse recovery.

S_1 turn-off with positive phase current

$0 < \alpha < \pi \qquad i_a > 0$

Ideal IGBT switches. Red: load path. Orange: recovery current.



$t < t_0$

S_1 conducts

t_0 to t_1

vCE_1 rises

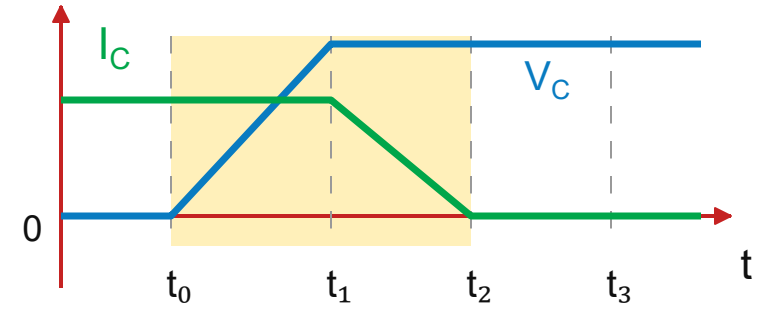
t_1 to t_2

Current transfers

$t > t_3$

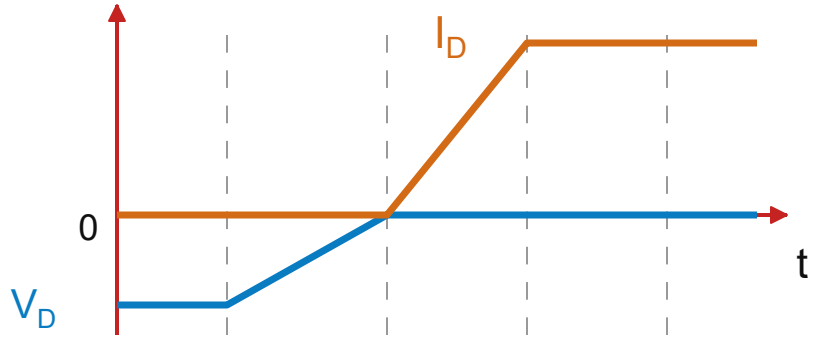
D_2 freewheels

S1 voltage and transistor current



Shaded: turn-off overlap

D2 voltage and current



No reverse recovery in this transition

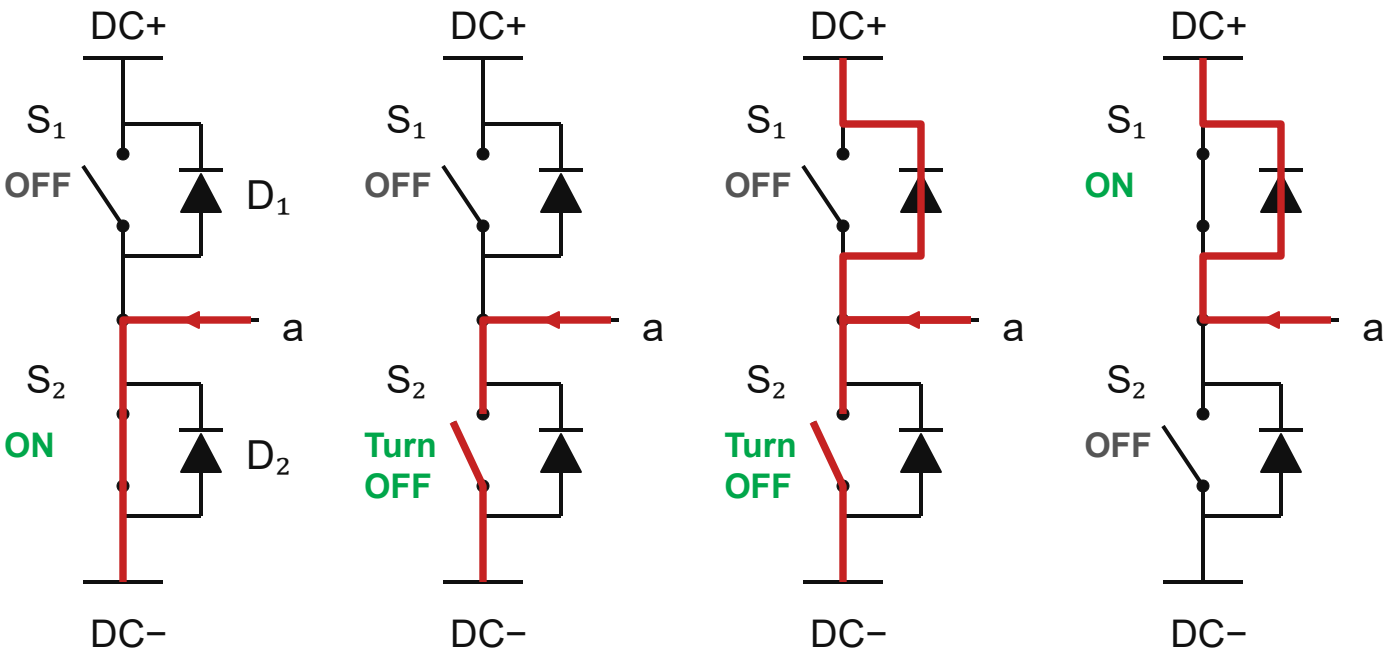
ES_1 = Eoff $\qquad ED_2,rr = 0$

S_2 receives ON at t_3. Positive current stays in D_2.

S_2 turn-off with negative phase current

$\pi < \alpha < 2\pi$ $\quad i_a < 0$

Ideal IGBT switches. Red: load path. Orange: recovery current.



$t < t_0$

S_2 conducts

t_0 to t_1

vCE_2 rises

t_1 to t_2

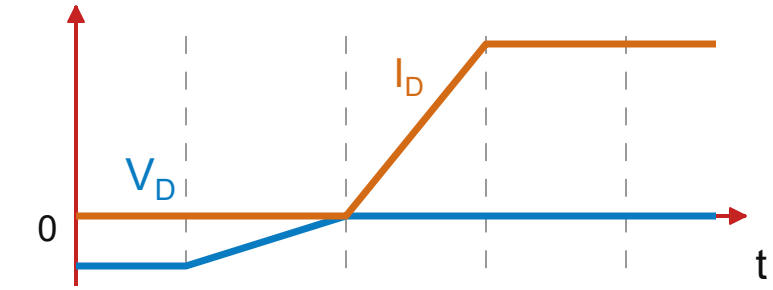
Current transfers

$t > t_3$

D_1 freewheels

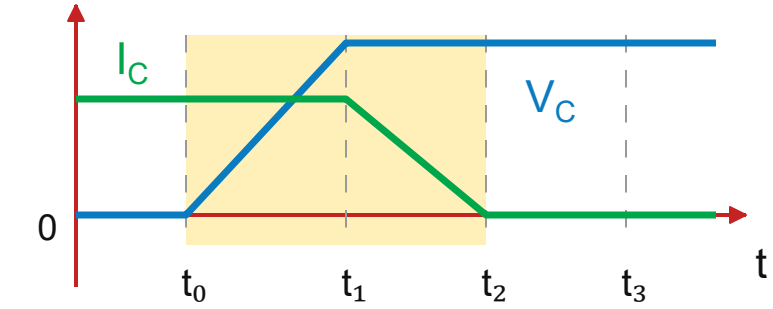
S_1 receives ON at t_3. Negative current stays in D_1.

D1 voltage and current



No reverse recovery in this transition

S2 voltage and transistor current



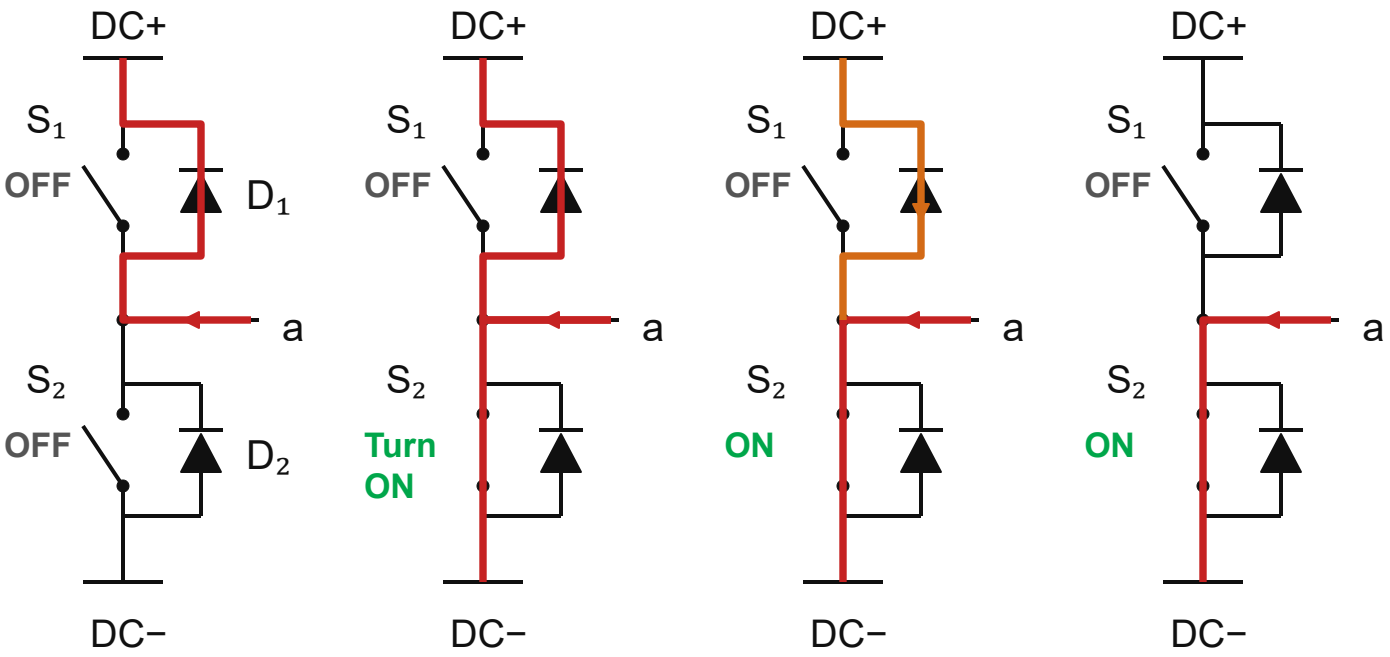
Shaded: turn-off overlap

ES_2 = Eoff $\quad$ **$ED_1,rr = 0$**

S_2 turn-on with negative phase current

$\pi < \alpha < 2\pi$ $\quad i_a < 0$

Ideal IGBT switches. Red: load path. Orange: recovery current.



$t < t_1$

D_1 freewheels

t_1 to t_2

S_2 takes current

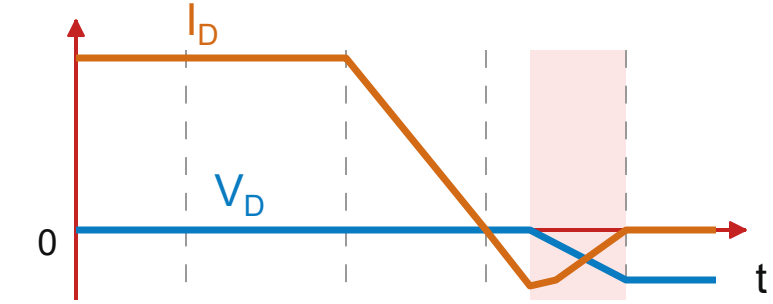
t_2 to t_3

D_1 recovers

$t > t_3$

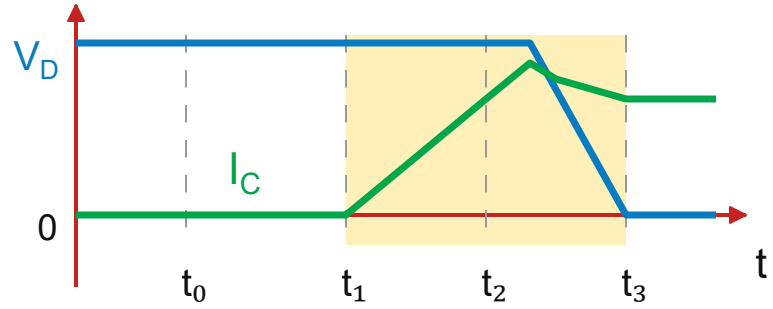
S_2 conducts

D1 voltage and current



Shaded: diode voltage–current overlap

S2 voltage and transistor current



Shaded: turn-on overlap

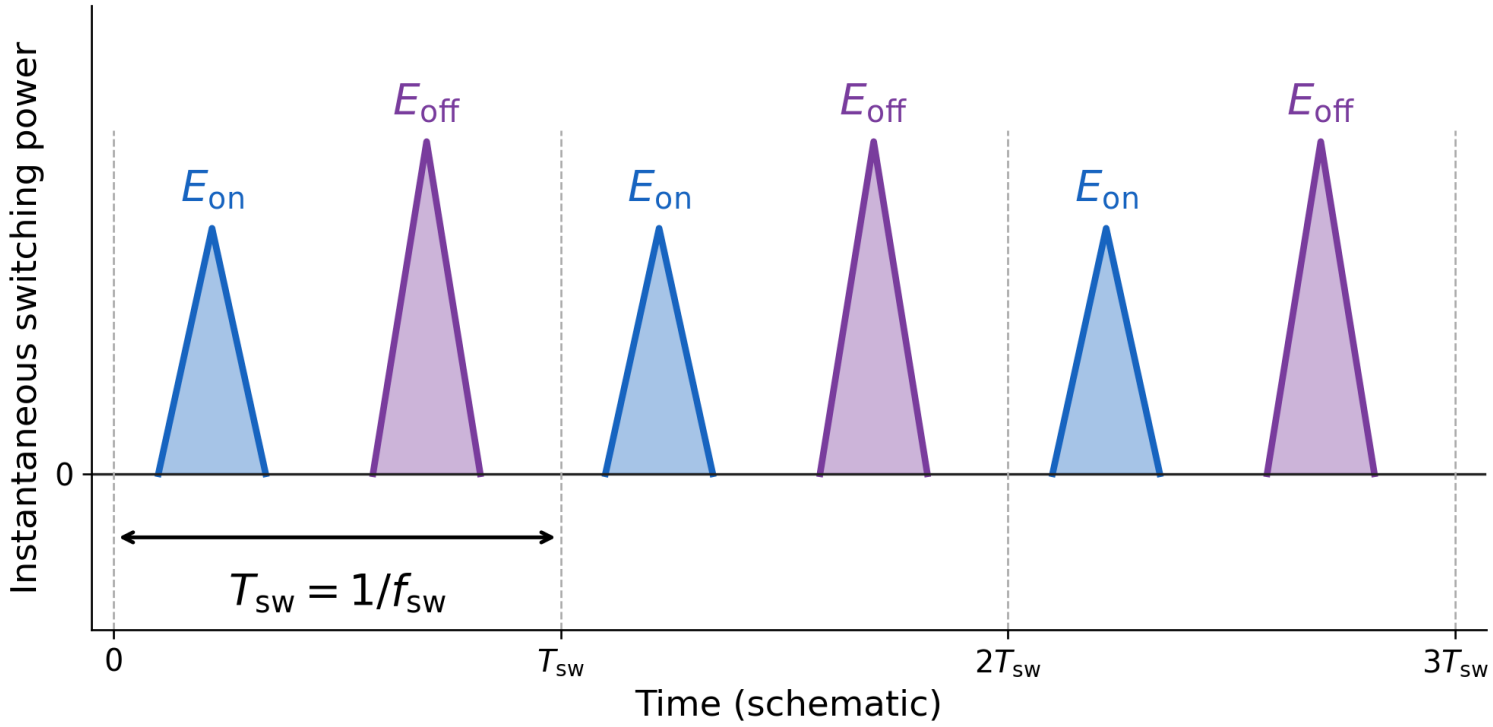
ES_2 = Eon $\quad$ **ED_1 = Err**

S_2 turns on and forces D_1 into reverse recovery.

If you have any confusion, refer back to this table

Device	Transition	First	Then
IGBT	Turn-on	Current rises	Voltage falls
IGBT	Turn-off	Voltage rises	Current falls
Diode	Turn-on	Voltage falls	Current rises
Diode	Turn-off	Forward current falls through zero and becomes negative	Reverse voltage rises, then reverse current decays to zero

From Switching Energy to Average Power Loss

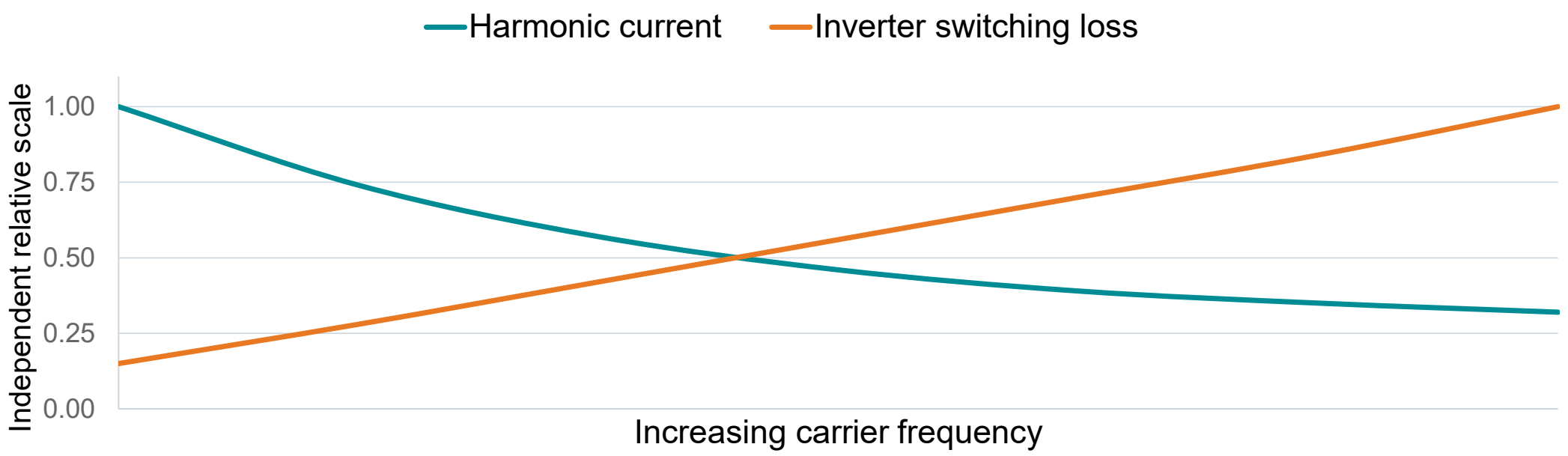


$$P_{sw} \approx f_{sw} \times (E_{on} + E_{off})$$

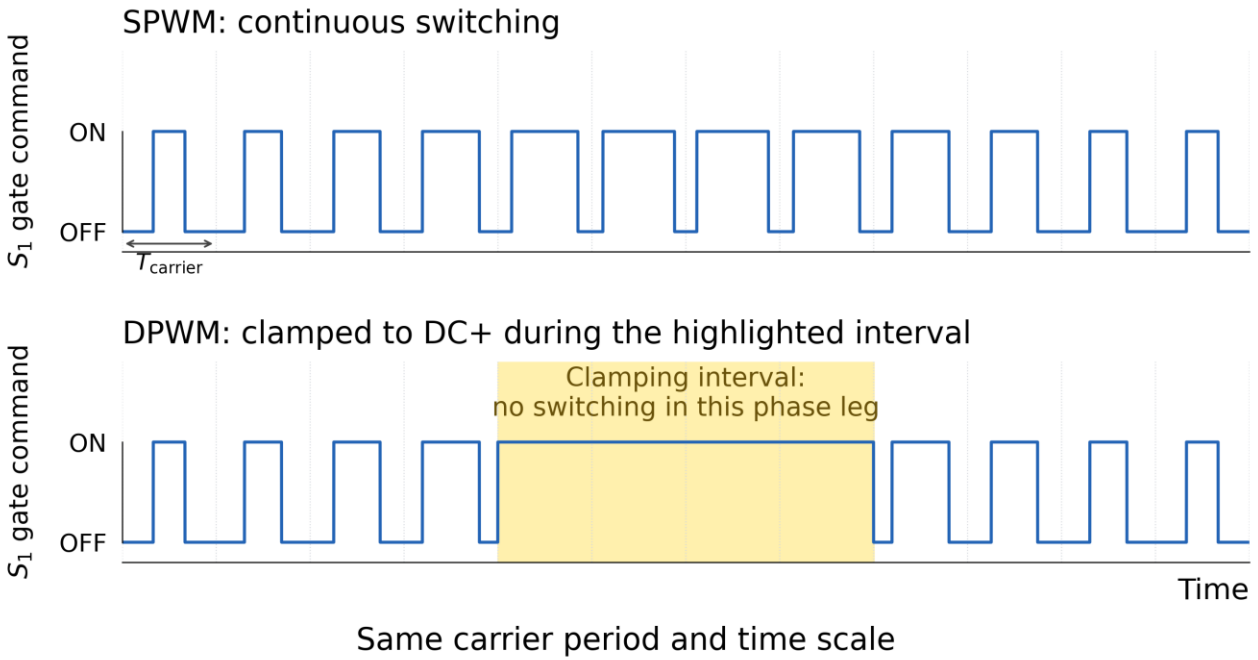
More switching events per second mean higher switching loss, if the energy per event stays the same.

The switching-frequency trade-off

- **Motor side:** higher frequency can reduce harmonic current through motor impedance.
- **Inverter side:** more switching events usually increase switching loss.
- **Actual trends** depend on the motor, device and operating point.



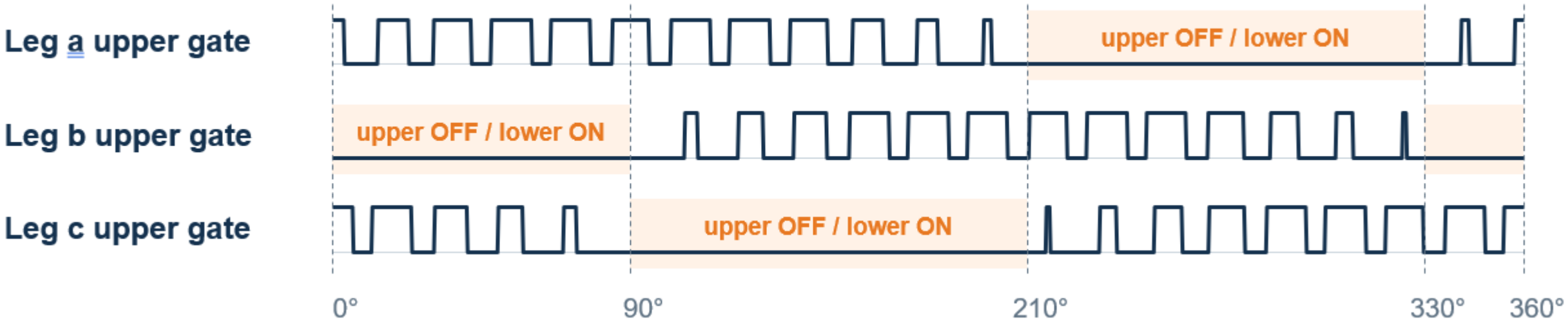
Modulation Strategy: Reducing Switching Events with DPWM



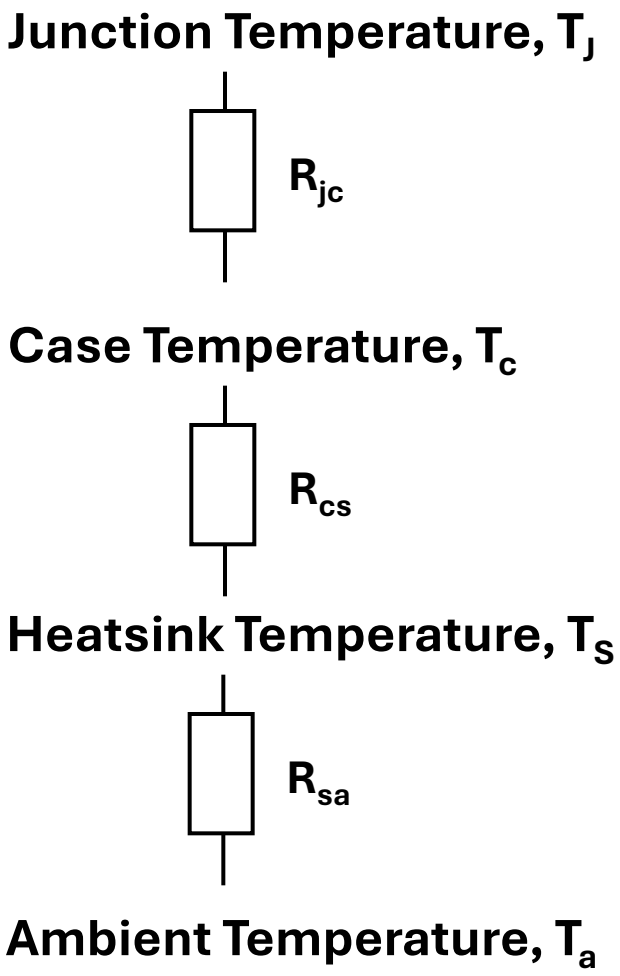
$$P_{sw} = \frac{1}{T_{obs}} \sum_{k} E_{sw,k}$$

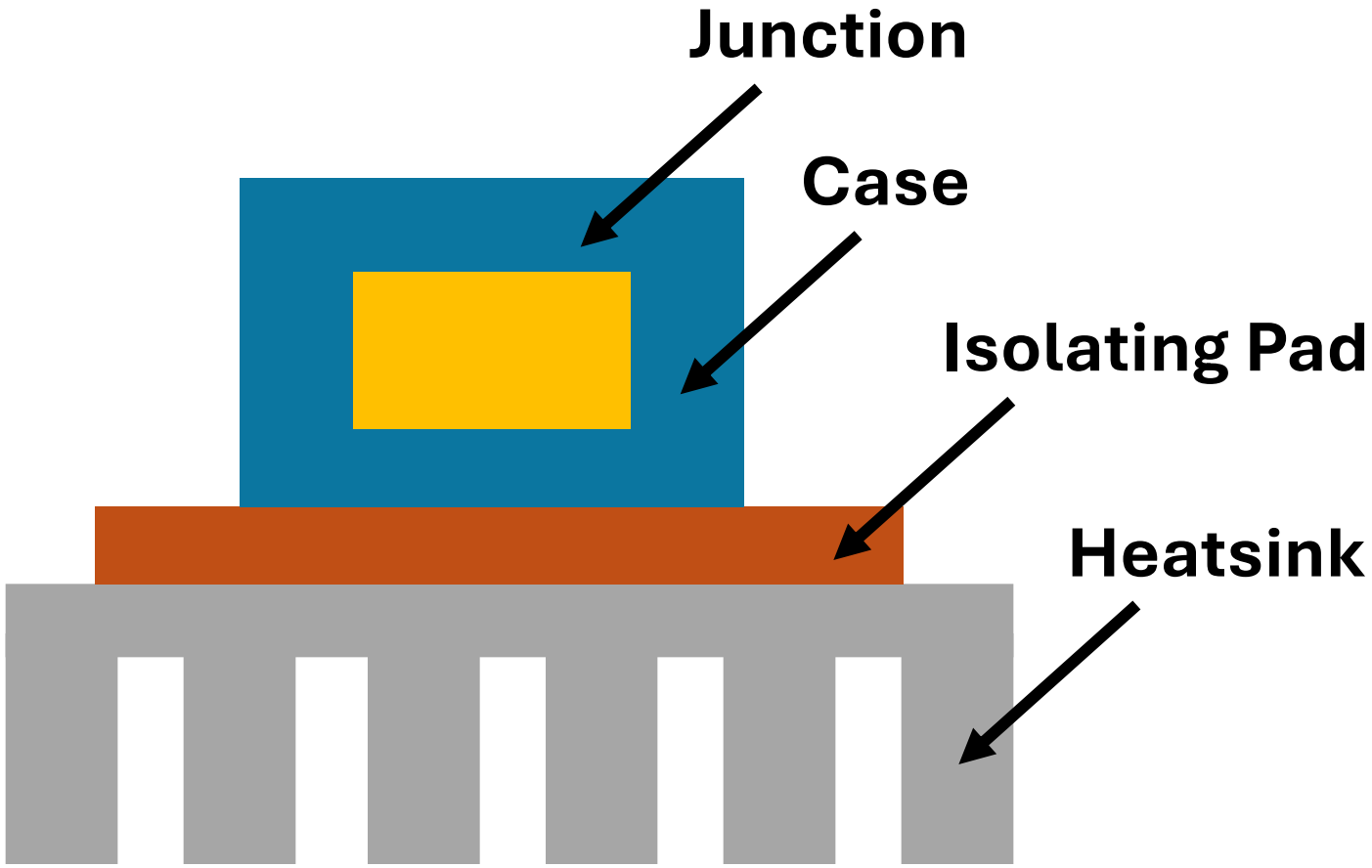
Schematic gate signals; not a complete modulation simulation

DPWMmin example: the lowest reference clamps its leg to negative rail O



Thermal Dissipation





$$\text{For Power Dissipation } T_j = P_d\,(R_{jc} + R_{cs} + R_{sa}) + T_a$$

Heat Dissipation Technologies

